

Francisco Ribeiro

Software engineer building backend systems, data pipelines and LLM-powered features.

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Porto, Portugal

Experience

Software Engineer · **Confidencial Imobiliário** Sept 2024 – Present

Full-time since Feb 2026, part-time before · Porto (hybrid)

- Work on the normalisation of every property and address that reaches the Portuguese market, combining several data sources into a single historical record per property.
- Built that pipeline in Django: LLMs parse free-text fields, rule-based merging decides which source wins by trust level, and geospatial comparison resolves addresses that no exact match can reach.
- Maintain and extend the company's CAS authentication server, fixing bugs, hardening it and adding third-party OAuth providers.
- Contributed to the subscription and service pipelines, including the Stripe integration for billing and plan management.
- Implemented the Django backend of the company's public website.
- Work daily with PostGIS, Celery, Redis and DRF APIs serving internal analytics teams.

Software Developer · **Ad Evolutio** Mar – June 2024

Internship · Porto

- Developed web components in Python/Django for a Geo Investment Platform (GeoIP), a tool for presenting a city's investment profile.
- Built interactive geospatial views with GeoServer and Leaflet over municipal open data.

Selected Project

MSc dissertation: **A Multi-Agent Framework Powered by Language Models for Real Estate Intelligence Analysis**

FEUP & Confidencial Imobiliário · 2025 – 2026

- Translates natural-language questions into spatial and temporal data operations and returns grounded SQL, maps, charts and narratives.
- Built on structured shared state, tool use, database grounding, review mechanisms and targeted fine-tuning of small open models.
- Designed the evaluation pipeline and benchmark used to measure the system end to end.
- Concluded that small open models can run the full analytical workflow inside such a system, and that system design reduces their limits without removing them.

Skills

Languages

Python, JavaScript, Java, SQL

AI & Data

LangGraph, LLM agents and fine-tuning, RAG and information retrieval, NLP, evaluation pipelines and benchmarking, pandas

Backend

Django, Django REST Framework, Celery, Redis, PostgreSQL, REST APIs

Geospatial

PostGIS, GeoServer, Leaflet, spatial and temporal data modelling

Auth & Integrations

CAS single sign-on, OAuth 2.0, Stripe

Infrastructure

Docker, Git, Linux, Google Cloud Platform

Education

Faculty of Engineering, University of Porto (FEUP)

MSc in Informatics and Computing Engineering
2024 – 2026

Software architecture, distributed systems, machine learning, NLP, computer vision.

Faculty of Sciences, University of Porto (FCUP)

BSc in Computer Science
2021 – 2024

Algorithms, databases, security, data science.

Also

Department Coordinator

NUCC – Núcleo de Estudantes de Ciência de Computadores, FCUP
2022 – 2024

Interests

Cycling, maps, geospatial data, nature, graphic design and photography.

Languages

Portuguese (native)
English (professional working proficiency)